2020 AMC 10B Problem 24 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 24. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 24

Problem:

How many positive integers n satisfy

$$\frac{n+1000}{70} = \lfloor \sqrt{n} \rfloor?$$

(Recall that $\lfloor x \rfloor$ is the greatest integer not exceeding x.)

Answer Choices:

A. 2
B. 4
C. 6
D. 30
E. 32

Solution:

We can first consider the equation without a floor function:

$$\frac{n+1000}{70} = \sqrt{n}$$

Multiplying both sides by 70 and then squaring:

$$n^2 + 2000n + 1000000 = 4900n$$

Moving all terms to the left:

$$n^2 - 2900n + 1000000 = 0$$

Now we can determine the factors:

$$(n - 400)(n - 2500) = 0$$

This means that for $n = 400$ and $n = 2500$, the equation will hold without the floor function. Now we can simply check the multiples of 70 around 400 and 2500 in the original equation:

- For $n = 330$, left hand side $= 19$ but $18^2 < 330 < 19^2$ so right hand side $= 18$
- For $n = 400$, left hand side $= 20$ and right hand side $= 20$
- For $n = 470$, left hand side $= 21$ and right hand side $= 21$
- For $n = 540$, left hand side $= 22$ but $540 > 23^2$ so right hand side $= 23$

Now we move to $n = 2500$

- For $n = 2430$, left hand side $= 49$ and $49^2 < 2430 < 50^2$ so right hand side $= 49$
- For $n = 2360$, left hand side $= 48$ and $48^2 < 2360 < 49^2$ so right hand side $= 48$
- For $n = 2290$, left hand side $= 47$ and $47^2 < 2360 < 48^2$ so right hand side $= 47$
- For $n = 2220$, left hand side $= 46$ but $47^2 < 2220$ so right hand side $= 47$
- For $n = 2500$, left hand side $= 50$ and right hand side $= 50$
- For $n = 2570$, left hand side $= 51$ but $2570 < 51^2$ so right hand side $= 50$

Therefore we have $\textbf{(C)} \boxed{6}$ total solutions, $n = 400, 470, 2290, 2360, 2430, 2500$.

OR

We are given that

$$\frac{n+1000}{70} = \lfloor \sqrt{n} \rfloor$$

$\lfloor \sqrt{n} \rfloor$ must be an integer, which means that $n+1000$ is divisible by 70 . As $1000 \equiv 20 (\bmod 70)$, this means that $n \equiv 50 (\bmod 70)$, so we can write $n = 70k + 50$ for $k \in \mathbb{Z}$. Therefore,

$$\frac{n+1000}{70} = \frac{70k+1050}{70} = k + 15 = \lfloor \sqrt{70k+50} \rfloor$$

Also, we can say that $\sqrt{70k+50} - 1 < k + 15$ and $k + 15 \leq \sqrt{70k+50}$ Squaring the second inequality, we get

$$k^2 + 30k + 225 \leq 70k + 50 \implies k^2 - 40k + 175 \leq 0 \implies (k-5)(k-35) \leq 0 \implies 5 \leq k \leq 35$$

Similarly solving the first inequality gives us

$$k < 19 - \sqrt{155} \text{ or } k > 19 + \sqrt{155}$$

$\sqrt{155}$ is larger than 12 and smaller than 13, so instead, we can say $k \leq 6$ or $k \geq 32$. Combining this with $5 \leq k \leq 35$, we get

$$k = 5, 6, 32, 33, 34, 35$$

are all solutions for k that give a valid solution for n, meaning that our answer is $\textbf{(C)} \boxed{6}$.